

5. Zinc reacts with sulfuric acid,  $\text{H}_2\text{SO}_4$ , to produce zinc sulfate and hydrogen gas.

(a) Write the formula of zinc sulfate to complete the equation for this reaction. [1]



(b) Some Year 10 students decided to investigate the rate of this reaction at 20 °C.

They added 0.2g of zinc to 50 cm<sup>3</sup> of sulfuric acid and measured the volume of hydrogen gas produced every 20 seconds for 160 seconds using a gas syringe.

Their results are shown in the table.

| Time (s) | Volume of hydrogen (cm <sup>3</sup> ) |
|----------|---------------------------------------|
| 0        | 0                                     |
| 20       | 20                                    |
| 40       | 37                                    |
| 60       | 50                                    |
| 80       | 60                                    |
| 100      | 66                                    |
| 120      | 69                                    |
| 140      | 70                                    |
| 160      | 70                                    |

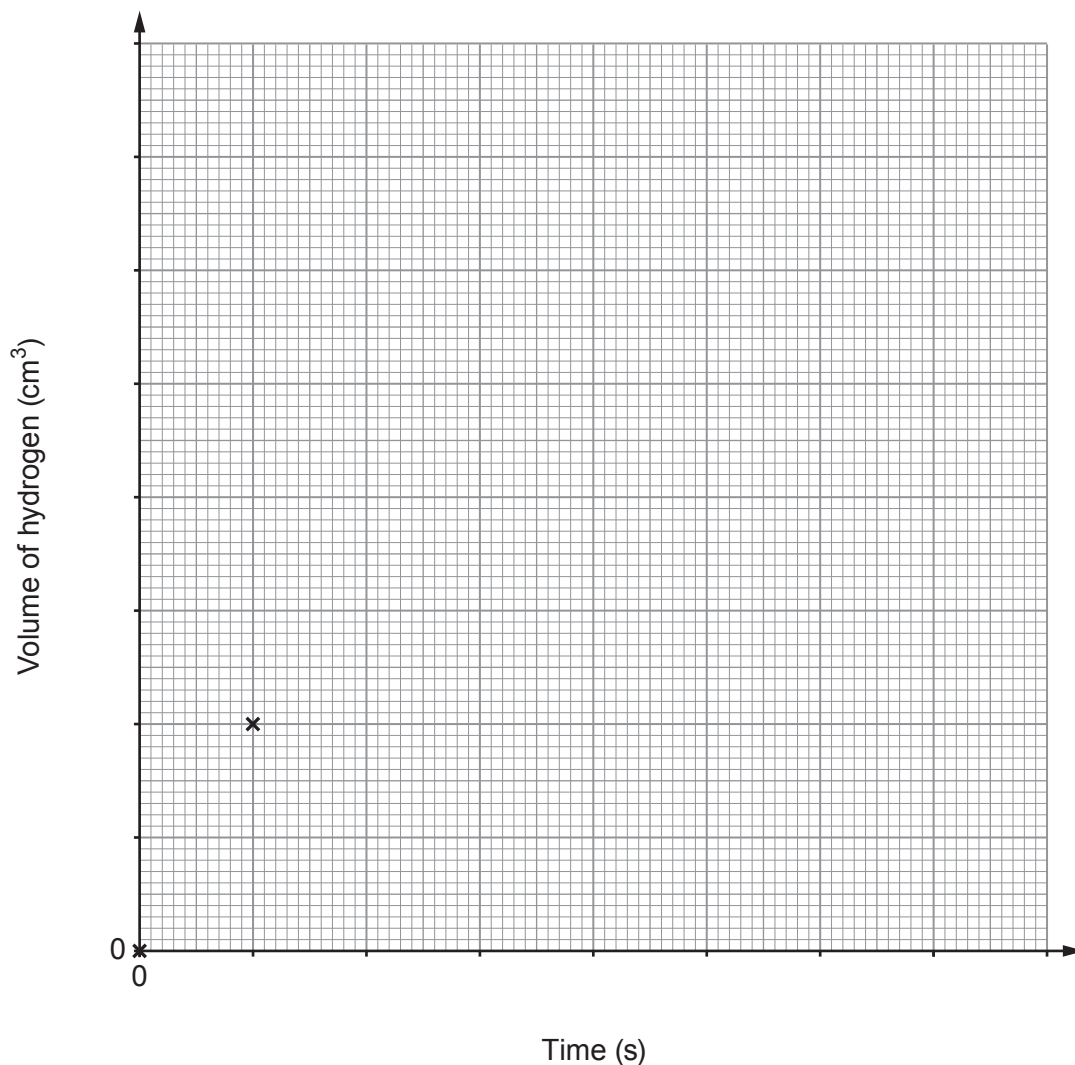


Draw appropriate scales on both axes on the grid.

Plot the volume of hydrogen produced against time and draw a suitable line.

[3]

The first two points have been plotted for you.



- (c) **On the same grid**, sketch the graph you would expect to obtain if the experiment were repeated at 40°C using the same mass of zinc and the same volume and concentration of sulfuric acid.

[1]



- (d) Use the particle theory to explain what you would expect to happen to the rate of the reaction if the experiment were repeated at 20 °C using sulfuric acid of **lower concentration**. [3]

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- (e) Copper is a useful catalyst in the reaction between zinc and sulfuric acid.

State how a catalyst increases the rate of a reaction. [1]

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